

MATHEMATICAL MODELLING OF CONTROL SYSTEMS

The control theory that we deal with here is the linear control theory, where the systems that are under consideration are **Dynamic**, **Linear**, and **Time Invariant** systems. A dynamic system, if the cause and effect are proportional, it is called linear. In a linear system, the principle of superposition applies. That is, the response to several inputs can be calculated by treating one input at a time and adding the results. Such systems will be governed by linear differential equations. A differential equation is linear if the coefficients are constants (time invariant) or functions only of the independent variable (time varying).

Transfer Function

The transfer function of a linear, time-invariant system is defined as the ratio of the Laplace transform of the output (Response function) to the Laplace transform of the input (Driving function) under the assumption that all initial conditions are zero.

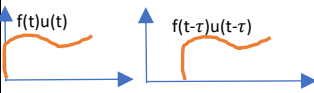
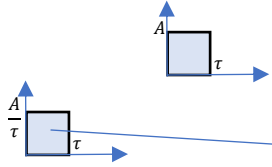
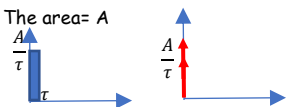
$$\text{Transfer Function} = G(s) = \left\{ \frac{\mathcal{L}[\text{output}]}{\mathcal{L}[\text{input}]} \right\}_{\text{zero initial conditions}}$$

- The transfer function of a system is a mathematical model which is an operational method of expressing the differential equation that relates the output variable to the input variable.
- The transfer function is a property of a system itself, independent of the magnitude and nature of the input or driving function.
- TF does not provide any information concerning the physical structure of the system. The transfer functions of many physically different systems can be identical.
- If the transfer function of a system is known, the output response can be studied for several inputs for understanding the nature of the system.
- If the transfer function of a system is unknown, it may be established experimentally by introducing known inputs and studying the output of the system. Once established, the transfer function gives a full description of the dynamic characteristics of the system.

A revisit to Laplace Transform

$$\mathcal{L}[f(t)] = F(s) = \int_0^{\infty} f(t) e^{-st} dt$$

Step function	$f(t) = A, \text{ for } t \geq 0$	$\mathcal{L}[f(t)] = \int_0^{\infty} A e^{-st} dt = A \left[\frac{e^{-st}}{-s} \right]_0^{\infty} = \frac{A}{s}$
	If $A = 1$, it is unit step function $u(t)$	$\mathcal{L}[u(t)] = \frac{1}{s}$
Ramp function	$f(t) = At, \text{ for } t \geq 0$	$\mathcal{L}[f(t)] = \int_0^{\infty} At e^{-st} dt = A t \left[\frac{e^{-st}}{-s} \right]_0^{\infty} - A \int_0^{\infty} 1 \cdot \frac{e^{-st}}{-s} dt = \frac{A}{s} \left[\frac{e^{-st}}{-s} \right]_0^{\infty} = \frac{A}{s^2}$ $\mathcal{L}[t^n] = \frac{n!}{s^{n+1}}$
Harmonic function	$f(t) = \sin \omega t$	$\mathcal{L}[\sin \omega t] = \int_0^{\infty} \frac{e^{i\omega t} - e^{-i\omega t}}{2i} e^{-st} dt = -\frac{i}{2} \left\{ \int_0^{\infty} e^{-(s-i\omega)t} dt - \int_0^{\infty} e^{-(s+i\omega)t} dt \right\}$ $= -\frac{i}{2} \left\{ \left[\frac{e^{-(s-i\omega)t}}{-(s-i\omega)} \right]_0^{\infty} - \left[\frac{e^{-(s+i\omega)t}}{-(s+i\omega)} \right]_0^{\infty} \right\} = -\frac{i}{2} \left\{ \left[\frac{1}{(s-i\omega)} \right] - \left[\frac{1}{(s+i\omega)} \right] \right\}$ $= -\frac{i}{2} \left[\frac{(s+i\omega) - (s-i\omega)}{s^2 + \omega^2} \right] = -\frac{i}{2} \left[\frac{2i\omega}{s^2 + \omega^2} \right] = \frac{\omega}{s^2 + \omega^2}$
	$f(t) = \cos \omega t$	$\mathcal{L}[\cos \omega t] = \frac{s}{s^2 + \omega^2}$

Translated function		$\mathcal{L}[f(t - \alpha)] = \int_0^{\infty} f(t - \tau) u(t - \tau) e^{-st} dt$ put $t - \tau = \alpha \gg dt = d\alpha$ $\int_0^{\infty} f(t - \tau) u(t - \tau) e^{-st} dt = \int_0^{\infty} f(\alpha) u(\alpha) e^{-s(\alpha + \tau)} d\alpha$ $= e^{-s\tau} \int_0^{\infty} f(\alpha) u(\alpha) e^{-s\alpha} d\alpha = e^{-s\tau} F(s)$
Pulse function	$f(t) = A, \text{ for } 0 \leq t \leq \tau$ $= A u(t) - A u(t - \tau)$ 	$\mathcal{L}[f(t)] = \int_0^{\infty} A \cdot u(t) e^{-st} dt - \int_0^{\infty} A u(t - \tau) e^{-st} dt$ $= \frac{A}{s} - e^{-s\tau} \frac{A}{s} = \frac{A}{s} (1 - e^{-s\tau})$ The strength of the pulse is the area of rectangle. $\mathcal{L}[f(t)] = \frac{A}{s} (1 - e^{-s\tau})$
Impulse	$g(t) = \lim_{\tau \rightarrow 0} \frac{A}{\tau} : 0 \leq t \leq \tau$ As $\tau \rightarrow 0$, the height $\frac{A}{\tau} \rightarrow \infty$ The area = A 	$\mathcal{L}[g(t)] = \lim_{\tau \rightarrow 0} \left\{ \frac{A}{s} (1 - e^{-s\tau}) \right\} = \lim_{\tau \rightarrow 0} \left\{ \frac{A s e^{-s\tau}}{s} \right\} = A$ If $A = 1$, it is unit impulse function (dirac delta function $\delta(t)$) $\mathcal{L}[\delta(t)] = 1$
Multiplication by e^{-at}	$e^{-at} f(t)$	$\mathcal{L}[e^{-at} f(t)] = \int_0^{\infty} e^{-at} f(t) e^{-st} dt = \int_0^{\infty} f(t) e^{-(s+a)t} dt = F(s + a)$
Exponential function	$f(t) = e^{-at}$	$\mathcal{L}[f(t)] = \int_0^{\infty} e^{-at} u(t) e^{-st} dt = \frac{1}{s + a}$
Real differential theorem	$\mathcal{L}\left[\frac{d}{dt} f(t)\right]$	$\int_0^{\infty} f(t) e^{-st} dt = \left[f(t) \frac{e^{-st}}{-s} \right]_0^{\infty} - \int_0^{\infty} f'(t) \frac{e^{-st}}{-s} dt$ $F(s) = \frac{f(0)}{s} + \frac{1}{s} \mathcal{L}[f'(t)] \gg \mathcal{L}[f'(t)] = sF(s) - f(0)$
	$\mathcal{L}\left[\frac{d^n}{dt^n} f(t)\right]$	$\mathcal{L}\left[\frac{d^n}{dt^n} f(t)\right] = s^n F(s) - s^{n-1} f(0) - s^{n-2} f'(0) - \dots$
Real Integration theorem	$\mathcal{L}\left[\int f(t) dt\right]$	$\int_0^{\infty} \left[\int f(t) dt \right] e^{-st} dt = \left[\int f(t) dt \frac{e^{-st}}{-s} \right]_0^{\infty} - \int_0^{\infty} f(t) \frac{e^{-st}}{-s} dt$ $\mathcal{L}\left[\int f(t) dt\right] = \left[\int \frac{f(t) dt}{s} \right]_{t=0}^{\infty} + \frac{1}{s} F(s) = \frac{F(s)}{s} + \frac{f^{-1}(0)}{s}$
Final value theorem If $f(t)$ and $f'(t)$ are Laplace transformable, if $F(s)$ is the Laplace transform of $f(t)$, and $\lim_{t \rightarrow \infty} f(t)$ exists, then $\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s)$ $\mathcal{L}[f'(t)] = sF(s) - f(0) \quad \ll \gg \quad \int_0^{\infty} \left[\frac{d}{dt} f(t) \right] e^{-st} dt = sF(s) - f(0)$ $\lim_{s \rightarrow 0} \int_0^{\infty} \left[\frac{d}{dt} f(t) \right] e^{-st} dt = \lim_{s \rightarrow 0} \{sF(s) - f(0)\}$ $\lim_{s \rightarrow 0} e^{-st} \rightarrow 1 \quad \ll \gg \quad [f(t)]_0^{\infty} = \lim_{s \rightarrow 0} \{sF(s)\} - f(0)$ $f(\infty) - f(0) = \lim_{s \rightarrow 0} \{sF(s)\} - f(0)$ $\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s)$		

Initial value theorem

If $f(t)$ and $f'(t)$ are both Laplace transformable, if $F(s)$ is the Laplace transform of $f(t)$, and $\lim_{s \rightarrow \infty} sF(s)$ exists, then $f(0^+) = \lim_{s \rightarrow \infty} sF(s)$

$$\mathcal{L}[f'(t)] = sF(s) - f(0^+) \quad \ll \gg \quad \int_0^{\infty} \left[\frac{d}{dt} f(t) \right] e^{-st} dt = sF(s) - f(0^+)$$

$$\lim_{s \rightarrow \infty} \int_0^{\infty} \left[\frac{d}{dt} f(t) \right] e^{-st} dt = \lim_{s \rightarrow \infty} \{sF(s) - f(0^+)\}$$

$$\lim_{s \rightarrow \infty} e^{-st} \rightarrow 0 \quad \ll \gg \quad 0 = \lim_{s \rightarrow \infty} \{sF(s)\} - f(0^+)$$

$$f(0^+) = \lim_{s \rightarrow \infty} \{sF(s)\}$$